

HEALTHCARE ANALYTICS & HOSPITAL PATIENT TREATMENT OUTCOME REPORT

Student Name: Mohd Sirajuddin Ahmed

Project Name: Hospital Patient Analytics & Treatment Outcome Dashboard

Tableau Public Link:

https://public.tableau.com/views/Healthcare_Tableau_Project_MohdSirajuddinAhmed/HospitalOverviewDashboard-1?:language=en-GB&:sid=&:redirect=auth&:display_count=n&:origin=viz_share_link

Dashboards Walk Through Video:

https://drive.google.com/file/d/1ayPFLYItQzgUtmn3HDKT1UQRE0B-t8y/_view?usp=drive_link

Project Overview & Business Scenario

Title: Hospital Patient Analytics & Treatment Outcome Dashboard

Role: Data Analyst / Business Analyst

Tools Used: Tableau Public, Data Visualization, Healthcare Analytics

Background & Scenario

Healthcare organizations generate massive volumes of patient and treatment data. Analyzing this data helps hospitals improve patient care outcomes, optimize overall treatment costs, and evaluate individual doctor and department performance.

As a Data Analyst, the objective was to analyze hospital patient data across Andhra Pradesh and Telangana to deliver executive-level interactive Tableau dashboards for management decision-making.

Core Objectives

- Analyze patient admission and discharge trends over time.
 - Measure key health outcomes including recovery rates, mortality rates, and readmission performance.
 - Evaluate disease severity against average treatment costs.
 - Assess overall doctor and department operational efficiency.
 - Develop interactive Tableau dashboards equipped with custom filters, drill-downs, and key performance indicators (KPIs).
-

Key Business Questions Addressed

1. Which diseases have the highest patient count?
 2. Which hospital departments are the most expensive to operate?
 3. What is the overall and disease-wise recovery rate?
 4. Which doctors show the best and worst performance based on patient outcomes and length of stay?
 5. Are less severe diseases treated at significantly lower costs compared to critical conditions?
 6. Which cities and states account for the highest volume of hospital admissions?
-

Calculated Metrics Implemented

Length of Stay: DATEDIFF('day', [Admission_Date], [Discharge_Date])

Recovery Rate: Percentage of patients with Recovery_Status = 'Recovered'

Mortality Rate: Percentage of patients with Recovery_Status = 'Expired'

Average Treatment Cost: Mean financial cost per treatment/patient

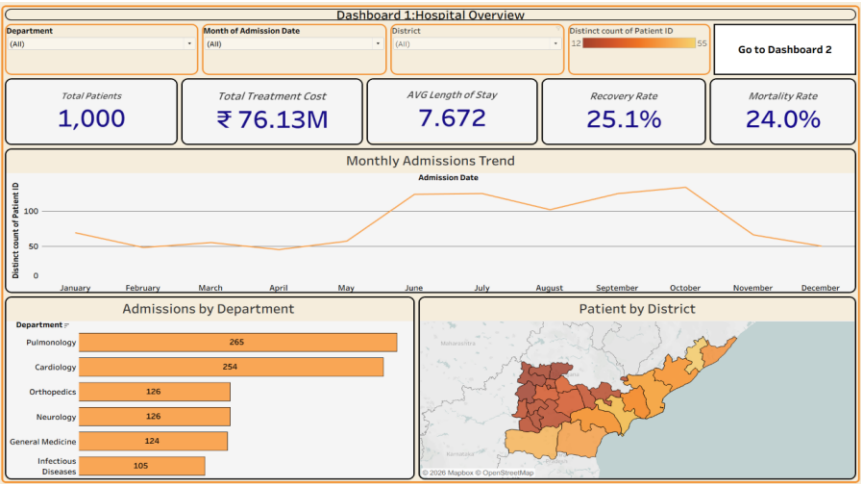
1. EXECUTIVE SUMMARY & PROJECT OBJECTIVES

This project analyzes patient admission trends, recovery rates, treatment costs, and doctor efficiency across regional hospital operations in Andhra Pradesh and Telangana. Using interactive visualization dashboards, key operational metrics were extracted to assist healthcare management in decision-making and resource

optimization.

2. DASHBOARD ANALYSIS & KEY INSIGHTS

Dashboard 1: Hospital Overview



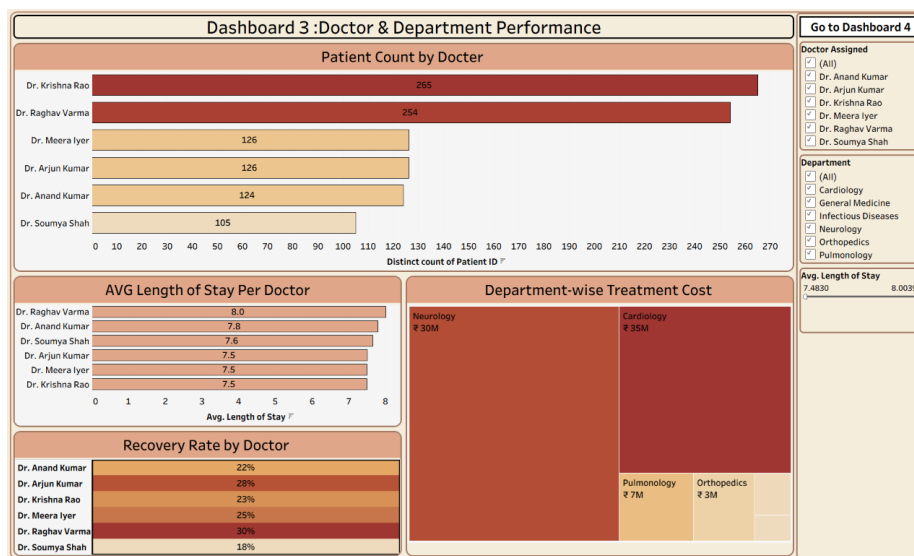
- Total Patient Volume: 1,000 admitted patients across all departments.
- Financial Metric: Total treatment cost stands at ₹76.13M, with an average length of stay of 7.67 days.
- Admission Trends: Admissions peak significantly during mid-year months (June–October).
- Departmental Load: Pulmonology (265) and Cardiology (254) represent the largest patient volume.

Dashboard 2: Disease Analysis



- High-Volume Conditions: Asthma (141) and Heart Attack (128) record the highest patient admissions.
- Severity & Cost: Heart Attack (₹273K avg) and Stroke (₹240K avg) demand the highest average treatment costs, whereas non-critical infections remain low-cost.
- Outcome Variance: Recovery rates vary across conditions, with Hypertension achieving the highest recovery rate at 37%.

Dashboard 3: Doctor & Department Performance

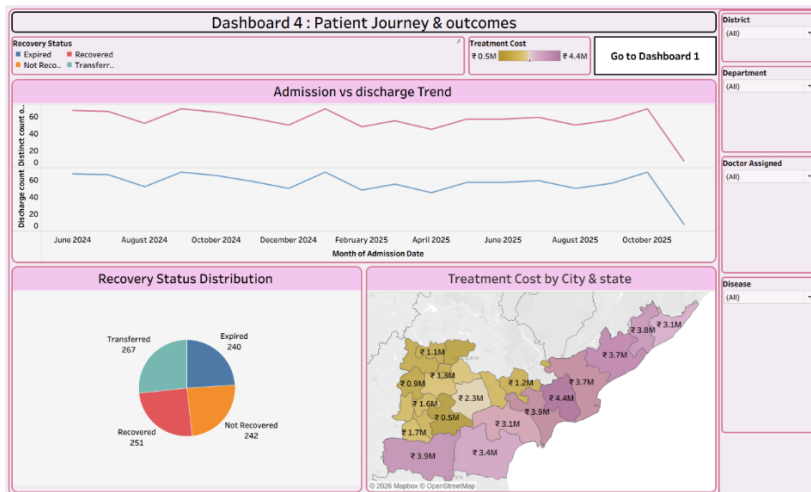


- Patient Load: Dr. Krishna Rao (265) and Dr. Raghav Varma (254) manage the highest

patient cases.

- Treatment Cost Distribution: Cardiology (₹35M) and Neurology (₹30M) drive over 80% of total hospital treatment costs.
- Efficiency Metrics: Length of stay per doctor remains consistent (7.5 to 8.0 days), showing standardized inpatient care timelines.

Dashboard 4: Patient Journey & Outcomes



- Outcome Breakdown: Distributed across Recovered (251), Not Recovered (242), Transferred (267), and Expired (240).
- Regional Distribution: Geographic analysis across Andhra Pradesh and Telangana shows high cost concentration in major district hubs.
- Admissions vs. Discharges: Stable monthly patient turnover with matching trends between admission and discharge volumes.

3. BUSINESS PROBLEMS & SOLUTIONS

1. Which diseases have the highest patient count?

-> Asthma (141) and Heart Attack (128).

2. Which departments are most expensive?

-> Cardiology (₹35M) and Neurology (₹30M).

3. What is the overall and disease-wise recovery rate?

-> Overall recovery rate is 25.1%, ranging from 18% (Infection) to 37% (Hypertension).

4. Which doctors show the best and worst performance?

-> Dr. Raghav Varma achieved the highest recovery rate (30%), while Dr. Soumya Shah recorded lower relative recovery rates (18%).

5. Are less severe diseases treated at lower cost compared to critical diseases?

-> Yes, conditions like Infection (₹8K) and Hypertension (₹4K) cost significantly less than critical cases like Heart Attack (₹273K).

6. Which cities/districts have the highest hospital admissions?

-> High concentrations are observed across primary district hospital centers in AP and TS.

4. STRATEGIC BUSINESS RECOMMENDATIONS

1. Resource Allocation: The data shows patient admissions increase from June to October.

The hospital should prepare extra beds and assign more staff to Pulmonology and Cardiology during these peak months to manage patient traffic easily.

2. Cost Control: Treatment costs for Cardiology and Neurology are very high. The hospital management should review treatment plans for these departments to reduce extra expenses without affecting treatment quality.

3. Improving Patient Outcomes: Recovery rates vary across different doctors. Doctors with high recovery rates can share their best treatment practices with others so that overall patient care becomes more consistent.

4. Better Regional Facilities: A high number of patients are being transferred from district centers. Upgrading medical facilities at small regional centers will help treat patients locally and prevent unnecessary emergency transfers.